Rapport detaille — NLS copy

Classification · Hold-out x 5 · Plateforme tfmv1 · 24 octobre 2026 (Europe/Madrid)

Rapport détaillé — NLS copy (classification)

Contexte commun

Champ	Valeur
Plateforme	http://46.62.212.226 (tfmv1)
Utilisateur	cc@cc.cc
Dataset	nls_copy (NLS copy)
Cible	class (5 classes: 0.0 … 4.0)
Split plateforme	train 125973 / test 22544 / total 148517
Validation	Hold-out répété × 5, test_size = 0.2 (interne au train)
Évaluation UI / finale	métriques sur le test set global (22544 lignes)
Features	~119 colonnes numériques (one-hot services/flags inclus)
Date du run	24 octobre 2026

Tableau récapitulatif

#	Algori...	Début (...	Fin (Ma...	Durée	Acc. ho...	Acc. test	Precision	Recall	F1	Train ...
1	Logistic ...	2026-10...	2026-10...	711.1s	0.9893±...	0.7611	0.7538	0.7611	0.7229	0.9900
2	k-Neares...	2026-10...	2026-10...	284.1s	0.9921±...	0.7304	0.7209	0.7304	0.6805	0.9954
3	LightGBM	2026-10...	2026-10...	17.5s	0.9970±...	0.7588	0.7635	0.7588	0.7274	0.9983
4	Random ...	2026-10...	2026-10...	31.4s	0.9988±...	0.7440	0.8044	0.7440	0.6972	0.9999
5	Neural N...	2026-10...	2026-10...	748.6s	0.9785±...	0.7114	0.6694	0.7114	0.6682	0.9796
6	Support ...	2026-10...	2026-10...	4710.2s	0.9921±...	0.7665	0.8074	0.7665	0.7315	0.9929

1. Logistic Regression

Identité & timing

Champ	Valeur
Algorithm (code)	logistic / model_type loggé logistic
Training ID	logistic_1792824556_f6646334
Dataset	nls_copy
User ID	6a7649c85389ffb49703be24
Status	success / event training_finished
Started (UTC)	2026-10-24T06:49:16.723545+00:00
Started (Europe/Madrid)	2026-10-24 08:49:16 (Europe/Madrid)
Finished (UTC)	2026-10-24T07:01:07.802810+00:00
Finished (Europe/Madrid)	2026-10-24 09:01:07 (Europe/Madrid)
Duration	711.079 s (11.85 min)
Result timestamp	2026-10-24T07:01:07.557951

Requête / paramètres envoyés (start log)

Paramètre request	Valeur
C_search_space	None
bootstrap_params	None

class_weight	None
cv_params	None
date_column	None
l1_ratio	None
max_iter	1000
model_type	logistic
multi_class	auto
n_repetitions	5
n_trials	20
numerized_dataset_name	nls_copy
original_dataset_name	nls_copy
penalty	l2
random_search	False
random_search_seed	42
random_state	None
solver	lbfgs
time_series_params	None
training_id	logistic_1792824556_f6646334
use_scaling	True
validation_method	holdout

Méthode de validation

Champ	Valeur
Méthode	Hold-Out Repeated (5 repetitions)
n_repetitions	5
test_size	0.2
mean_score (holdout)	0.989268
std_score (holdout)	0.000546
best_repetition	5
best_score	0.989919
best_C (agrégé)	1000.0

Détail de chaque répétition

Rep	Score validation	Score train	train_size	val_size	test%	best_C
1	0.989680	0.989680	100778	25195	—	100.000000
2	0.989125	0.990008	100778	25195	—	100.000000
3	0.988331	0.990087	100778	25195	—	1000.000000
4	0.989284	0.989849	100778	25195	—	100.000000
5	0.989919	0.989978	100778	25195	—	1000.000000

Recherche de C par répétition (Logistic)

Répétition 1 — best_C=100.0, score=0.989680

C	Score validation
0.001	0.970232
0.01	0.978964
0.1	0.988688
1	0.989006
10	0.989165
100	0.989680
1000	0.989561

Répétition 2 — best_C=100.0, score=0.989125

C	Score validation
0.001	0.970073
0.01	0.978647
0.1	0.987617
1	0.988569
10	0.988847
100	0.989125
1000	0.989045

Répétition 3 — best_C=1000.0, score=0.988331

C	Score validation
0.001	0.970073
0.01	0.978448
0.1	0.986902
1	0.988013
10	0.988172
100	0.988212
1000	0.988331

Répétition 4 — best_C=100.0, score=0.989284

C	Score validation
0.001	0.970669
0.01	0.978845
0.1	0.988093
1	0.988887
10	0.989244
100	0.989284
1000	0.989006

Répétition 5 — best_C=1000.0, score=0.989919

C	Score validation
0.001	0.969438
0.01	0.977813
0.1	0.988252
1	0.989601
10	0.989760
100	0.989760
1000	0.989919

Hyperparamètres & paramètres effectifs


```
{
  "metadata": {
    "model_type": "logistic_regression",
    "C_used": 1000.0,
    "C_selection_method": "validation_split",
    "penalty": "l2",
    "solver": "lbfgs",
    "multi_class": "auto",
    "n_features": 119,
    "n_train_samples": 125973,
    "n_test_samples": 22544,
    "validation_method": "holdout",
    "class_names": [
      "0.0",
      "1.0",
      "2.0",
```

```
      "3.0",
      "4.0"
    ],
    "used_validation_model": true,
    "final_model_source": "best_holdout_repetition_5",
    "date_column_excluded": false
  },
  "best_C_used": 1000.0,
  "C_search_space": [
    0.001,
    0.01,
    0.1,
    1,
    10,
    100,
    1000
  ],
  "C_selection_method": "validation_split",
  "training_info": {
    "model_trained_once": false,
    "used_cv_best_model": false,
    "used_bootstrap_model": false,
    "used_holdout_model": true,
    "used_timeseries_model": false,
    "final_model_source": "best_holdout_repetition_5",
    "training_samples": 125973,
    "test_samples": 22544,
    "parameter_optimization": "C optimization using holdout",
    "n_repetitions": 5,
    "validation_method": "holdout",
    "date_column_excluded": false,
    "date_column": null
  }
}
```


Features utilisées (119):

flag_sh, duration, protocol_type, src_bytes, dst_bytes, land, wrong_fragment, urgent, hot, num_failed_logins, logged_in, num_compromised, root_shell, su_attempted, num_root, num_file_creations, num_shells, num_access_files, is_host_login, is_guest_login, count, srv_count, serror_rate, srv_serror_rate, rerror_rate, srv_rerror_rate, same_srv_rate, diff_srv_rate, srv_diff_host_rate, dst_host_count, dst_host_srv_count, dst_host_same_srv_rate, dst_host_diff_srv_rate, dst_host_same_src_port_rate, dst_host_srv_diff_host_rate, dst_host_serror_rate, dst_host_srv_serror_rate, dst_host_rerror_rate, dst_host_srv_rerror_rate, service_aol, service_auth, service_bgp, service_courier, service_csnet_ns, service_ctf, service_daytime, service_discard, service_domain, service_domain_u, service_echo, service_eco_i, service_ecr_i, service_efs, service_exec, service_finger, service_ftp, service_ftp_data, service_gopher, service_harvest, service_hostnames, service_http, service_http_2784, service_http_443, service_http_8001, service_imap4, service_irc, service_iso_tsap, service_klogin, service_kshell, service_ldap, service_link, service_login, service_mtp, service_name, service_netbios_dgm, service_netbios_ns, service_netbios_ssn, service_netstat, service_nnsp, service_nntp, service_ntp_u, service_other, service_pm_dump, service_pop_2, service_pop_3, service_printer, service_private, service_red_i, service_remote_job, service_rje, service_shell, service_smtp, service_sql_net, service_ssh, service_sunrpc, service_supdup, service_systat, service_telnet, service_tftp_u, service_tim_i, service_time, service_urh_i, service_urp_i, service_uucp, service_uucp_path, service_vmnet, service_whois, service_x11, service_z39_50, flag_oth, flag_rej, flag_rsto, flag_rstos0, flag_rstr, flag_s0, flag_s1, flag_s2, flag_s3, flag_sf

Métriques finales (affichage UI / test set)

Métrique	Valeur
accuracy	0.761089
precision	0.753803
recall	0.761089
f1	0.722912
roc_auc	None
log_loss	2.842005
train_accuracy	0.989966
final_test.accuracy	0.761089
final_test.precision	0.753803
final_test.recall	0.761089
final_test.f1	0.722912
final_train.accuracy	0.989966
n_classes	5
classes	['0.0', '1.0', '2.0', '3.0', '4.0']
target_column	class

Matrice de confusion (test)

actual \ pred	0.0	1.0	2.0	3.0	4.0
0.0	6223	1233	4	0	0
1.0	79	9032	595	1	4
2.0	118	425	1749	129	0
3.0	3	2741	5	128	8
4.0	2	35	0	4	26

Classification report (test)

Classe / avg	precision	recall	f1-score	support
0.0	0.968560	0.834182	0.896363	7460.0
1.0	0.670726	0.930079	0.779393	9711.0
2.0	0.743306	0.722429	0.732719	2421.0
3.0	0.488550	0.044367	0.081347	2885.0
4.0	0.684211	0.388060	0.495238	67.0
accuracy			0.761089	
macro avg	0.711071	0.583823	0.597012	22544.0
weighted avg	0.753803	0.761089	0.722912	22544.0

Feature importance (top 20)

Rang	Feature	Importance
1	is_guest_login	5.291409
2	num_file_creations	5.025860
3	duration	3.999612
4	hot	3.904614
5	protocol_type	2.671285
6	wrong_fragment	2.420069
7	dst_host_same_srv_rate	2.275400
8	count	2.016701
9	srv_count	1.945216
10	service_private	1.636728
11	same_srv_rate	1.620601

12	num_root	1.557458
13	service_eco_i	1.488315
14	dst_host_serror_rate	1.410893
15	flag_sf	1.402888
16	serror_rate	1.392022
17	flag_s0	1.321427
18	srv_serror_rate	1.267117
19	num_compromised	1.187302
20	dst_host_srv_count	1.132245

training_info


```
{
  "model_trained_once": false,
  "used_cv_best_model": false,
  "used_bootstrap_model": false,
  "used_holdout_model": true,
  "used_timeseries_model": false,
  "final_model_source": "best_holdout_repetition_5",
  "training_samples": 125973,
  "test_samples": 22544,
  "parameter_optimization": "C optimization using holdout",
  "n_repetitions": 5,
  "validation_method": "holdout",
  "date_column_excluded": false,
  "date_column": null
}
```


2. k-Nearest Neighbors (KNN)

Identité & timing

Champ	Valeur
Algorithm (code)	knn / model_type loggé knn
Training ID	knn_1792825269_9b68403d
Dataset	nls_copy
User ID	6a7649c85389ffb49703be24
Status	success / event training_finished
Started (UTC)	2026-10-24T07:01:09.332073+00:00
Started (Europe/Madrid)	2026-10-24 09:01:09 (Europe/Madrid)
Finished (UTC)	2026-10-24T07:05:53.406878+00:00
Finished (Europe/Madrid)	2026-10-24 09:05:53 (Europe/Madrid)
Duration	284.075 s (4.73 min)
Result timestamp	2026-10-24T07:05:53.034722

Requête / paramètres envoyés (start log)

Paramètre request	Valeur
algorithm	auto
bootstrap_params	None
cv_params	None
final_training_on_all_data	False
leaf_size	30
metric	minkowski

model_type	knn
n_neighbors	5
n_repetitions	5
n_trials	20
numerized_dataset_name	nls_copy
original_dataset_name	nls_copy
p	2
random_search	False
random_search_seed	42
time_series_column	None
time_series_params	None
training_id	knn_1792825269_9b68403d
use_scaling	False
validation_method	holdout
weights	uniform

Méthode de validation

Champ	Valeur
Méthode	Hold-Out Repeated (5 repetitions)
n_repetitions	5
test_size	0.2
mean_score (holdout)	0.992141
std_score (holdout)	0.000466
best_repetition	5
best_score	0.992538

Détail de chaque répétition

Rep	Score validation	Score train	train_size	val_size	test%
1	0.992300	0.994632	100778	25195	20.0
2	0.992260	0.994592	100778	25195	20.0
3	0.992379	0.994791	100778	25195	20.0
4	0.991228	0.994840	100778	25195	20.0
5	0.992538	0.994642	100778	25195	20.0

Hyperparamètres & paramètres effectifs


```
{
  "n_neighbors_used": 5,
  "weights_used": "uniform",
  "metric_used": "minkowski",
  "algorithm_used": "auto",
  "final_params_used": {
    "n_neighbors": 5,
    "weights": "uniform",
    "algorithm": "auto",
    "leaf_size": 30,
    "p": 2,
    "metric": "minkowski"
  },
  "training_info": {
    "final_training_data": "train_only",
    "all_data_samples": 125973,
    "train_samples_used": 125973,
    "test_samples_used": 0,
    "final_model_trained_on": "train_data_only",
    "n_repetitions": 5,
    "validation_method": "holdout",
    "date_column_excluded": false,
```

```
    "scaling_applied": false,
    "model_trained_once": false,
    "used_cv_best_model": false,
    "used_bootstrap_model": false,
    "used_holdout_repeated_model": true,
    "used_timeseries_model": false,
    "final_model_source": "best_holdout_fold_5",
    "training_samples": 125973,
    "test_samples": 22544,
    "feature_scaling": "No scaling applied",
    "parameter_optimization": "User parameters used - no optimization"
  }
}
```


Features utilisées (119):

duration, protocol_type, src_bytes, dst_bytes, land, wrong_fragment, urgent, hot, num_failed_logins, logged_in, num_compromised, root_shell, su_attempted, num_root, num_file_creations, num_shells, num_access_files, is_host_login, is_guest_login, count, srv_count, serror_rate, srv_serror_rate, rerror_rate, srv_rerror_rate, same_srv_rate, diff_srv_rate, srv_diff_host_rate, dst_host_count, dst_host_srv_count, dst_host_same_srv_rate, dst_host_diff_srv_rate, dst_host_same_src_port_rate, dst_host_srv_diff_host_rate, dst_host_serror_rate, dst_host_srv_serror_rate, dst_host_rerror_rate, dst_host_srv_rerror_rate, service_aol, service_auth, service_bgp, service_courier, service_csnet_ns, service_ctf, service_daytime, service_discard, service_domain, service_domain_u, service_echo, service_eco_i, service_ecr_i, service_efs, service_exec, service_finger, service_ftp, service_ftp_data, service_gopher, service_harvest, service_hostnames, service_http, service_http_2784, service_http_443, service_http_8001, service_imap4, service_irc, service_iso_tsap, service_klogin, service_kshell, service_ldap, service_link, service_login, service_mtp, service_name, service_netbios_dgm, service_netbios_ns, service_netbios_ssn, service_netstat, service_nnsp, service_nntp, service_ntp_u, service_other, service_pm_dump, service_pop_2, service_pop_3, service_printer, service_private, service_red_i, service_remote_job, service_rje, service_shell, service_smtp, service_sql_net, service_ssh, service_sunrpc, service_supdup, service_systat, service_telnet, service_tftp_u, service_tim_i, service_time, service_urh_i, service_urp_i, service_uucp, service_uucp_path, service_vmnet, service_whois, service_x11, service_z39_50, flag_oth, flag_rej, flag_rsto, flag_rstos0, flag_rstr, flag_s0, flag_s1, flag_s2, flag_s3, flag_sf, flag_sh

Métriques finales (affichage UI / test set)

Métrique	Valeur
accuracy	0.730438
precision	0.720892
recall	0.730438
f1	0.680477
roc_auc	None
train_accuracy	0.995436
final_test.accuracy	0.730438
final_test.precision	0.720892
final_test.recall	0.730438
final_test.f1	0.680477
final_train.accuracy	0.995436
n_classes	5
classes	['0.0', '1.0', '2.0', '3.0', '4.0']
target_column	class
effective_n_neighbors	5

Matrice de confusion (test)

actual \ pred	0.0	1.0	2.0	3.0	4.0
0.0	5565	1602	293	0	0
1.0	131	9438	134	8	0
2.0	290	681	1450	0	0
3.0	52	2680	147	6	0
4.0	0	59	0	0	8

Classification report (test)

Classe / avg	precision	recall	f1-score	support
0.0	0.921663	0.745979	0.824567	7460.0
1.0	0.652697	0.971888	0.780936	9711.0
2.0	0.716403	0.598926	0.652418	2421.0
3.0	0.428571	0.002080	0.004139	2885.0
4.0	1.000000	0.119403	0.213333	67.0
accuracy			0.730438	
macro avg	0.743867	0.487655	0.495079	22544.0
weighted avg	0.720892	0.730438	0.680477	22544.0

Feature importance (top 20)

Rang	Feature	Importance
1	src_bytes	0.593282
2	dst_bytes	0.406408
3	duration	0.000263
4	count	0.000012
5	dst_host_srv_count	0.000011
6	dst_host_count	0.000010
7	srv_count	0.000007
8	num_root	0.000002
9	num_compromised	0.000002
10	hot	0.000000
11	flag_sf	0.000000
12	logged_in	0.000000
13	num_file_creations	0.000000
14	service_http	0.000000
15	dst_host_same_srv_rate	0.000000
16	flag_s0	0.000000
17	srv_serror_rate	0.000000
18	serror_rate	0.000000
19	dst_host_srv_serror_rate	0.000000
20	dst_host_serror_rate	0.000000

training_info


```
{
  "final_training_data": "train_only",
  "all_data_samples": 125973,
  "train_samples_used": 125973,
  "test_samples_used": 0,
  "final_model_trained_on": "train_data_only",
  "n_repetitions": 5,
  "validation_method": "holdout",
  "date_column_excluded": false,
  "scaling_applied": false,
  "model_trained_once": false,
```

```
  "used_cv_best_model": false,
  "used_bootstrap_model": false,
  "used_holdout_repeated_model": true,
  "used_timeseries_model": false,
  "final_model_source": "best_holdout_fold_5",
  "training_samples": 125973,
  "test_samples": 22544,
  "feature_scaling": "No scaling applied",
  "parameter_optimization": "User parameters used - no optimization"
}
```


3. LightGBM

Identité & timing

Champ	Valeur
Algorithm (code)	lightgbm / model_type loggé lightgbm
Training ID	lightgbm_1792825556_1083687e
Dataset	nls_copy
User ID	6a7649c85389ffb49703be24
Status	success / event training_finished
Started (UTC)	2026-10-24T07:05:56.898682+00:00
Started (Europe/Madrid)	2026-10-24 09:05:56 (Europe/Madrid)
Finished (UTC)	2026-10-24T07:06:14.367334+00:00
Finished (Europe/Madrid)	2026-10-24 09:06:14 (Europe/Madrid)
Duration	17.469 s (0.29 min)
Result timestamp	2026-10-24T07:06:14.150870

Requête / paramètres envoyés (start log)

Paramètre request	Valeur
boosting_type	gbdt
bootstrap_params	None
colsample_bytree	1.0
cv_params	None
early_stopping_rounds	10
final_training_on_all_data	False
learning_rate	0.1
max_depth	-1
min_child_samples	20
model_type	lightgbm
n_estimators	100
n_jobs	-1
n_repetitions	5
n_trials	20
num_leaves	31
numerized_dataset_name	nls_copy
objective	None
original_dataset_name	nls_copy
random_search	False
random_search_seed	42
random_state	42
reg_alpha	0.0

reg_lambda	0.0
subsample	1.0
time_series_column	None
time_series_params	None
training_id	lightgbm_1792825556_1083687e
validation_method	holdout
verbose	-1

Méthode de validation

Champ	Valeur
Méthode	Hold-Out Repeated (5 repetitions)
n_repetitions	5
test_size	0.2
mean_score (holdout)	0.997023
std_score (holdout)	0.000470
best_repetition	5
best_score	0.997539

Détail de chaque répétition

Rep	Score validation	Score train	train_size	val_size	test%	best_iteration
1	0.997500	0.998571	100778	25195	20.0	22
2	0.996984	0.998621	100778	25195	20.0	33
3	0.996825	0.998105	100778	25195	20.0	24
4	0.996269	0.998234	100778	25195	20.0	27
5	0.997539	0.998690	100778	25195	20.0	24

Hyperparamètres & paramètres effectifs


```
{
  "objective_used": "multiclass",
  "boosting_type_used": "gbdt",
  "n_estimators_used": 100,
  "learning_rate_used": 0.1,
  "final_params_used": {
    "boosting_type": "gbdt",
    "num_leaves": 31,
    "learning_rate": 0.1,
    "n_estimators": 100,
    "max_depth": -1,
    "min_child_samples": 20,
    "subsample": 1.0,
    "colsample_bytree": 1.0,
    "reg_alpha": 0.0,
    "reg_lambda": 0.0,
    "random_state": 42,
    "n_jobs": -1,
    "verbose": -1,
    "objective": "multiclass",
    "num_class": 5,
    "early_stopping_rounds": 10
  },
  "training_info": {
    "final_training_data": "train_only",
    "all_data_samples": 125973,
    "train_samples_used": 125973,
    "test_samples_used": 0,
    "final_model_trained_on": "train_data_only",
    "n_repetitions": 5,
    "validation_method": "holdout",
```

```
    "date_column_excluded": false,
    "early_stopping_used": true,
    "best_iteration": 0,
    "model_trained_once": false,
    "used_cv_best_model": false,
    "used_bootstrap_model": false,
    "used_holdout_repeated_model": true,
    "used_timeseries_model": false,
    "final_model_source": "best_holdout_fold_5",
    "training_samples": 125973,
    "test_samples": 22544,
    "parameter_optimization": "User parameters used - no optimization"
  }
}
```


Features utilisées (119):

duration, protocol_type, src_bytes, dst_bytes, land, wrong_fragment, urgent, hot, num_failed_logins, logged_in, num_compromised, root_shell, su_attempted, num_root, num_file_creations, num_shells, num_access_files, is_host_login, is_guest_login, count, srv_count, serror_rate, srv_serror_rate, rerror_rate, srv_rerror_rate, same_srv_rate, diff_srv_rate, srv_diff_host_rate, dst_host_count, dst_host_srv_count, dst_host_same_srv_rate, dst_host_diff_srv_rate, dst_host_same_src_port_rate, dst_host_srv_diff_host_rate, dst_host_serror_rate, dst_host_srv_serror_rate, dst_host_rerror_rate, dst_host_srv_rerror_rate, service_aol, service_auth, service_bgp, service_courier, service_csnet_ns, service_ctf, service_daytime, service_discard, service_domain, service_domain_u, service_echo, service_eco_i, service_ecr_i, service_efs, service_exec, service_finger, service_ftp, service_ftp_data, service_gopher, service_harvest, service_hostnames, service_http, service_http_2784, service_http_443, service_http_8001, service_imap4, service_irc, service_iso_tsap, service_klogin, service_kshell, service_ldap, service_link, service_login, service_mtp, service_name, service_netbios_dgm, service_netbios_ns, service_netbios_ssn, service_netstat, service_nnsp, service_nntp, service_ntp_u, service_other, service_pm_dump, service_pop_2, service_pop_3, service_printer, service_private, service_red_i, service_remote_job, service_rje, service_shell, service_smtp, service_sql_net, service_ssh, service_sunrpc, service_supdup, service_systat, service_telnet, service_tftp_u, service_tim_i, service_time, service_urh_i, service_urp_i, service_uucp, service_uucp_path, service_vmnet, service_whois, service_x11, service_z39_50, flag_oth, flag_rej, flag_rsto, flag_rstos0, flag_rstr, flag_s0, flag_s1, flag_s2, flag_s3, flag_sf, flag_sh

Métriques finales (affichage UI / test set)

Métrique	Valeur
accuracy	0.758827
precision	0.763506
recall	0.758827
f1	0.727432
roc_auc	None
train_accuracy	0.998341
final_test.accuracy	0.758827
final_test.precision	0.763506
final_test.recall	0.758827
final_test.f1	0.727432
final_train.accuracy	0.998341
n_classes	5
classes	['0.0', '1.0', '2.0', '3.0', '4.0']
target_column	class
best_iteration	0

Matrice de confusion (test)

actual \ pred	0.0	1.0	2.0	3.0	4.0
0.0	6047	801	193	213	206
1.0	74	9422	201	4	10
2.0	163	798	1391	7	62
3.0	0	2625	7	244	9
4.0	0	55	0	9	3

Classification report (test)

Classe / avg	precision	recall	f1-score	support
0.0	0.962285	0.810590	0.879948	7460.0
1.0	0.687687	0.970240	0.804886	9711.0
2.0	0.776228	0.574556	0.660337	2421.0
3.0	0.511530	0.084575	0.145152	2885.0
4.0	0.010345	0.044776	0.016807	67.0
accuracy			0.758827	
macro avg	0.589615	0.496947	0.501426	22544.0
weighted avg	0.763506	0.758827	0.727432	22544.0

Feature importance (top 20)

Rang	Feature	Importance
1	src_bytes	0.153846
2	dst_host_count	0.077692
3	dst_bytes	0.065641
4	dst_host_srv_count	0.064615
5	dst_host_same_src_port_rate	0.048205
6	dst_host_same_srv_rate	0.042564
7	dst_host_diff_srv_rate	0.037179
8	dst_host_serror_rate	0.036667
9	count	0.030513
10	duration	0.029231
11	dst_host_rerror_rate	0.026667
12	dst_host_srv_diff_host_rate	0.024103
13	diff_srv_rate	0.021795
14	dst_host_srv_serror_rate	0.020513
15	service_http	0.020256
16	hot	0.020000
17	service_ftp_data	0.018462
18	wrong_fragment	0.017436
19	service_private	0.016923
20	srv_count	0.015641

training_info


```
{
  "final_training_data": "train_only",
  "all_data_samples": 125973,
  "train_samples_used": 125973,
  "test_samples_used": 0,
  "final_model_trained_on": "train_data_only",
  "n_repetitions": 5,
  "validation_method": "holdout",
  "date_column_excluded": false,
  "early_stopping_used": true,
  "best_iteration": 0,
```

```
  "model_trained_once": false,
  "used_cv_best_model": false,
  "used_bootstrap_model": false,
  "used_holdout_repeated_model": true,
  "used_timeseries_model": false,
  "final_model_source": "best_holdout_fold_5",
  "training_samples": 125973,
  "test_samples": 22544,
  "parameter_optimization": "User parameters used - no optimization"
}
```


4. Random Forest

Identité & timing

Champ	Valeur
Algorithm (code)	random_forest / model_type loggé random_forest
Training ID	random_forest_1792825577_516a6ea2
Dataset	nls_copy
User ID	6a7649c85389ffb49703be24
Status	success / event training_finished
Started (UTC)	2026-10-24T07:06:17.058532+00:00
Started (Europe/Madrid)	2026-10-24 09:06:17 (Europe/Madrid)
Finished (UTC)	2026-10-24T07:06:48.428198+00:00
Finished (Europe/Madrid)	2026-10-24 09:06:48 (Europe/Madrid)
Duration	31.370 s (0.52 min)
Result timestamp	2026-10-24T07:06:48.273604

Requête / paramètres envoyés (start log)

Paramètre request	Valeur
bootstrap	True
bootstrap_params	None
criterion	gini
cv_params	None
final_training_on_all_data	False
max_depth	None
max_features	sqrt
min_samples_leaf	1
min_samples_split	2
model_type	random_forest
n_estimators	100
n_jobs	-1
n_repetitions	5
n_trials	20
numerized_dataset_name	nls_copy
oob_score	False
original_dataset_name	nls_copy
random_search	False
random_search_seed	42
time_series_column	None
time_series_params	None
training_id	random_forest_1792825577_516a6ea2

validation_method	holdout

Méthode de validation

Champ	Valeur
Méthode	Hold-Out Repeated (5 repetitions)
n_repetitions	5
test_size	0.2
mean_score (holdout)	0.998833
std_score (holdout)	0.000203
best_repetition	1
best_score	0.999167

Détail de chaque répétition

Rep	Score validation	Score train	train_size	val_size	test%	random_state
1	0.999167	0.999970	100778	25195	20.0	0
2	0.998730	0.999960	100778	25195	20.0	1
3	0.998651	0.999970	100778	25195	20.0	2
4	0.998968	0.999931	100778	25195	20.0	3
5	0.998651	0.999960	100778	25195	20.0	4

Hyperparamètres & paramètres effectifs


```
{
  "final_params_used": {
    "n_estimators": 100,
    "max_depth": null,
    "min_samples_split": 2,
    "min_samples_leaf": 1,
    "max_features": "sqrt",
    "bootstrap": true,
    "oob_score": false,
    "n_jobs": -1,
    "random_state": null,
    "criterion": "gini"
  },
  "user_parameters_used": true,
  "training_info": {
    "final_training_data": "train_only",
    "all_data_samples": 125973,
    "train_samples_used": 125973,
    "test_samples_used": 0,
    "final_model_trained_on": "train_data_only",
    "n_repetitions": 5,
    "validation_method": "holdout",
    "date_column_excluded": false,
    "final_training_on_all_data": false,
    "training_samples": 125973,
    "test_samples": 22544,
    "final_model_samples": 125973
  }
}
```


Features utilisées (119):

duration, protocol_type, src_bytes, dst_bytes, land, wrong_fragment, urgent, hot, num_failed_logins, logged_in, num_compromised, root_shell, su_attempted, num_root, num_file_creations, num_shells, num_access_files, is_host_login, is_guest_login, count, srv_count, serror_rate, srv_serror_rate, rerror_rate, srv_rerror_rate, same_srv_rate, diff_srv_rate, srv_diff_host_rate, dst_host_count, dst_host_srv_count, dst_host_same_srv_rate, dst_host_diff_srv_rate, dst_host_same_src_port_rate, dst_host_srv_diff_host_rate,

dst_host_serror_rate, dst_host_srv_serror_rate, dst_host_rerror_rate, dst_host_srv_rerror_rate, service_aol, service_auth, service_bgp, service_courier, service_csnet_ns, service_ctf, service_daytime, service_discard, service_domain, service_domain_u, service_echo, service_eco_i, service_ecr_i, service_efs, service_exec, service_finger, service_ftp, service_ftp_data, service_gopher, service_harvest, service_hostnames, service_http, service_http_2784, service_http_443, service_http_8001, service_imap4, service_irc, service_iso_tsap, service_klogin, service_kshell, service_ldap, service_link, service_login, service_mtp, service_name, service_netbios_dgm, service_netbios_ns, service_netbios_ssn, service_netstat, service_nnsp, service_nntp, service_ntp_u, service_other, service_pm_dump, service_pop_2, service_pop_3, service_printer, service_private, service_red_i, service_remote_job, service_rje, service_shell, service_smtp, service_sql_net, service_ssh, service_sunrpc, service_supdup, service_systat, service_telnet, service_tftp_u, service_tim_i, service_time, service_urh_i, service_urp_i, service_uucp, service_uucp_path, service_vmnet, service_whois, service_x11, service_z39_50, flag_oth, flag_rej, flag_rsto, flag_rstos0, flag_rstr, flag_s0, flag_s1, flag_s2, flag_s3, flag_sf, flag_sh

Métriques finales (affichage UI / test set)

Métrique	Valeur
accuracy	0.744012
precision	0.804357
recall	0.744012
f1	0.697176
train_accuracy	0.999944
final_test.accuracy	0.744012
final_test.precision	0.804357
final_test.recall	0.744012
final_test.f1	0.697176
final_train.accuracy	0.999944
n_classes	5
classes	['0.0', '1.0', '2.0', '3.0', '4.0']
target_column	class

Matrice de confusion (test)

actual \ pred	0.0	1.0	2.0	3.0	4.0
0.0	5821	1544	95	0	0
1.0	67	9458	185	0	1
2.0	161	802	1458	0	0
3.0	0	2847	3	34	1
4.0	0	62	0	3	2

Classification report (test)

Classe / avg	precision	recall	f1-score	support
0.0	0.962308	0.780295	0.861796	7460.0
1.0	0.642833	0.973947	0.774484	9711.0
2.0	0.837450	0.602230	0.700625	2421.0
3.0	0.918919	0.011785	0.023272	2885.0
4.0	0.500000	0.029851	0.056338	67.0
accuracy			0.744012	
macro avg	0.772302	0.479622	0.483303	22544.0
weighted avg	0.804357	0.744012	0.697176	22544.0

Feature importance (top 20)

Rang	Feature	Importance
1	src_bytes	0.121809
2	dst_bytes	0.073984
3	count	0.061095
4	diff_srv_rate	0.060343
5	same_srv_rate	0.057296
6	flag_sf	0.049856
7	serror_rate	0.044551
8	dst_host_diff_srv_rate	0.041200
9	logged_in	0.036941
10	dst_host_srv_count	0.035836
11	dst_host_serror_rate	0.032275
12	dst_host_same_src_port_rate	0.032245
13	dst_host_srv_serror_rate	0.030660
14	srv_serror_rate	0.028885
15	dst_host_same_srv_rate	0.028851
16	protocol_type	0.025905
17	dst_host_count	0.024906
18	dst_host_srv_diff_host_rate	0.022539
19	flag_s0	0.020992
20	srv_count	0.017697

training_info


```
{
  "final_training_data": "train_only",
  "all_data_samples": 125973,
  "train_samples_used": 125973,
  "test_samples_used": 0,
  "final_model_trained_on": "train_data_only",
  "n_repetitions": 5,
  "validation_method": "holdout",
  "date_column_excluded": false,
  "final_training_on_all_data": false,
  "training_samples": 125973,
  "test_samples": 22544,
  "final_model_samples": 125973
}
```


5. Neural Net — MLP (PyTorch)

Identité & timing

Champ	Valeur
Algorithm (code)	mlp / model_type loggé mlp
Training ID	mlp_1792825613_6266dad8
Dataset	nls_copy
User ID	6a7649c85389ffb49703be24
Status	success / event training_finished
Started (UTC)	2026-10-24T07:06:53.501599+00:00
Started (Europe/Madrid)	2026-10-24 09:06:53 (Europe/Madrid)
Finished (UTC)	2026-10-24T07:19:22.101782+00:00

Finished (Europe/Madrid)	2026-10-24 09:19:22 (Europe/Madrid)
Duration	748.600 s (12.48 min)
Result timestamp	2026-10-24T07:19:21.829832

Requête / paramètres envoyés (start log)

Paramètre request	Valeur
activation	relu
batch_size	32
bootstrap_params	None
cv_params	None
dropout_rate	0.2
epochs	100
layers	[64, 32]
learning_rate	0.001
model_type	mlp
n_repetitions	5
numerized_dataset_name	nls_copy
original_dataset_name	nls_copy
patience	10
problem_type	auto
time_series_column	None
time_series_params	None
training_id	mlp_1792825613_6266dad8
use_scaling	True
validation_method	holdout
validation_split	0.2

Méthode de validation

Champ	Valeur
Méthode	Hold-Out Repeated (5 repetitions)
n_repetitions	5
test_size	0.2
mean_score (holdout)	0.978456
std_score (holdout)	0.001115
best_repetition	3
best_score	0.979599

Détail de chaque répétition

Rep	Score vali...	Score train	train_size	val_size	test%	best_val_...	n_classes	output_siz...
1	0.978686	—	100778	25195	20.0	0.925606	5	5
2	0.979202	—	100778	25195	20.0	0.925139	5	5
3	0.979599	—	100778	25195	20.0	0.924970	5	5
4	0.976384	—	100778	25195	20.0	0.926934	5	5
5	0.978408	—	100778	25195	20.0	0.924826	5	5

Paramètres utilisés par répétition (MLP)

• Rep 1: {"layers": [64, 32], "activation": "relu", "dropout_rate": 0.2, "learning_rate": 0.001, "batch_size": 32, "epochs": 100, "patience": 10} | best_val_loss=0.9256059306272085 | score=0.978686
• Rep 2: {"layers": [64, 32], "activation": "relu", "dropout_rate": 0.2, "learning_rate": 0.001, "batch_size": 32, "epochs": 100, "patience": 10} | best_val_loss=0.925139229859263 | score=0.979202
• Rep 3: {"layers": [64, 32], "activation": "relu", "dropout_rate": 0.2, "learning_rate": 0.001, "batch_size": 32, "epochs": 100, "patience": 10} | best_val_loss=0.9249702927469616 | score=0.979599

• Rep 4: {"layers": [64, 32], "activation": "relu", "dropout_rate": 0.2, "learning_rate": 0.001, "batch_size": 32, "epochs": 100, "patience": 10} | best_val_loss=0.9269337865468362 | score=0.976384
• Rep 5: {"layers": [64, 32], "activation": "relu", "dropout_rate": 0.2, "learning_rate": 0.001, "batch_size": 32, "epochs": 100, "patience": 10} | best_val_loss=0.9248263185472332 | score=0.978408

Early stopping / historique (MLP) — résumé

Rep	epochs run	best epoch (min ...	min val_loss	last train acc	last val acc
1	44	34	0.925606	0.979757	0.978686
2	19	9	0.925139	0.978269	0.979202
3	31	21	0.924970	0.979132	0.979599
4	25	15	0.926934	0.978656	0.976384
5	29	19	0.924826	0.978597	0.978408

Hyperparamètres & paramètres effectifs


```
{
  "layers_used": [
    64,
    32
  ],
  "activation_used": "relu",
  "dropout_rate_used": 0.2,
  "learning_rate_used": 0.001,
  "batch_size_used": 32,
  "training_info": {
    "model_trained_once": false,
    "used_cv_best_model": false,
    "used_bootstrap_model": false,
    "used_holdout_model": true,
    "used_timeseries_model": false,
    "training_samples": 125973,
    "test_samples": 22544,
    "parameter_optimization": "User parameters used - no optimization",
    "final_model_source": "best_holdout_repetition",
    "early_stopping_used": true,
    "training_id": "mlp_1792825613_6266dad8",
    "date_column_excluded": false
  }
}
```


Métriques finales (affichage UI / test set)

Métrique	Valeur
accuracy	0.711364
precision	0.669404
recall	0.711364
f1	0.668218
roc_auc	None
train_accuracy	0.979623
n_classes	5
classes	['0.0', '1.0', '2.0', '3.0', '4.0']
target_column	class

Matrice de confusion (test)

actual \ pred	0.0	1.0	2.0	3.0	4.0
0.0	5427	1870	100	63	0
1.0	450	9023	237	1	0

2.0	242	478	1557	144	0
3.0	7	2844	4	30	0
4.0	4	63	0	0	0

Classification report (test)

Classe / avg	precision	recall	f1-score	support
0.0	0.885318	0.727480	0.798675	7460.0
1.0	0.631951	0.929153	0.752261	9711.0
2.0	0.820337	0.643123	0.721000	2421.0
3.0	0.126050	0.010399	0.019212	2885.0
4.0	0.000000	0.000000	0.000000	67.0
accuracy			0.711364	
macro avg	0.492731	0.462031	0.458230	22544.0
weighted avg	0.669404	0.711364	0.668218	22544.0

training_info


```
{
  "model_trained_once": false,
  "used_cv_best_model": false,
  "used_bootstrap_model": false,
  "used_holdout_model": true,
  "used_timeseries_model": false,
  "training_samples": 125973,
  "test_samples": 22544,
  "parameter_optimization": "User parameters used - no optimization",
  "final_model_source": "best_holdout_repetition",
  "early_stopping_used": true,
  "training_id": "mlp_1792825613_6266dad8",
  "date_column_excluded": false
}
```


6. Support Vector Machine (SVM)

Identité & timing

Champ	Valeur
Algorithm (code)	svm / model_type loggé svm
Training ID	svm_1792826365_45f08639
Dataset	nls_copy
User ID	6a7649c85389ffb49703be24
Status	success / event training_finished
Started (UTC)	2026-10-24T07:19:25.469590+00:00
Started (Europe/Madrid)	2026-10-24 09:19:25 (Europe/Madrid)
Finished (UTC)	2026-10-24T08:37:55.684371+00:00
Finished (Europe/Madrid)	2026-10-24 10:37:55 (Europe/Madrid)
Duration	4710.215 s (78.50 min)
Result timestamp	2026-10-24T08:37:55.457878

Requête / paramètres envoyés (start log)

Paramètre request	Valeur
C	1.0

bootstrap_params	None
cv_params	None
degree	3
final_training_on_all_data	False
gamma	scale
kernel	rbf
model_type	svm
n_repetitions	5
n_trials	20
numerized_dataset_name	nls_copy
original_dataset_name	nls_copy
probability	False
random_search	False
random_search_seed	42
random_state	None
shrinking	True
time_series_column	None
time_series_params	None
training_id	svm_1792826365_45f08639
validation_method	holdout

Méthode de validation

Champ	Valeur
Méthode	Hold-Out Repeated (5 repetitions)
n_repetitions	5
test_size	0.2
mean_score (holdout)	0.992078
std_score (holdout)	0.000401
best_repetition	2
best_score	0.992697

Détail de chaque répétition

Rep	Score validation	Score train	train_size	val_size	test%
1	0.991903	0.992925	100778	25195	20.0
2	0.992697	0.992578	100778	25195	20.0
3	0.992141	0.993004	100778	25195	20.0
4	0.991467	0.993024	100778	25195	20.0
5	0.992181	0.992846	100778	25195	20.0

Hyperparamètres & paramètres effectifs


```
{
  "kernel_used": "rbf",
  "C_used": 1.0,
  "gamma_used": "scale",
  "degree_used": 3,
  "final_params_used": {
    "kernel": "rbf",
    "C": 1.0,
    "gamma": "scale",
    "degree": 3,
    "shrinking": true,
    "probability": false
  },
  "user_parameters_used": true,
  "training_info": {
    "final_training_data": "train_only",
```

```
    "all_data_samples": 125973,
    "train_samples_used": 125973,
    "test_samples_used": 0,
    "final_model_trained_on": "train_data_only",
    "n_repetitions": 5,
    "validation_method": "holdout",
    "date_column_excluded": false,
    "final_training_on_all_data": false,
    "training_samples": 125973,
    "test_samples": 22544,
    "final_model_samples": 125973,
    "feature_scaling": "StandardScaler applied",
    "parameter_optimization": "User parameters used - no optimization",
    "final_model_source": "best_validation_model"
  }
}
```


Features utilisées (119):

duration, protocol_type, src_bytes, dst_bytes, land, wrong_fragment, urgent, hot, num_failed_logins, logged_in, num_compromised, root_shell, su_attempted, num_root, num_file_creations, num_shells, num_access_files, is_host_login, is_guest_login, count, srv_count, serror_rate, srv_serror_rate, rerror_rate, srv_rerror_rate, same_srv_rate, diff_srv_rate, srv_diff_host_rate, dst_host_count, dst_host_srv_count, dst_host_same_srv_rate, dst_host_diff_srv_rate, dst_host_same_src_port_rate, dst_host_srv_diff_host_rate, dst_host_serror_rate, dst_host_srv_serror_rate, dst_host_rerror_rate, dst_host_srv_rerror_rate, service_aol, service_auth, service_bgp, service_courier, service_csnet_ns, service_ctf, service_daytime, service_discard, service_domain, service_domain_u, service_echo, service_eco_i, service_ecr_i, service_efs, service_exec, service_finger, service_ftp, service_ftp_data, service_gopher, service_harvest, service_hostnames, service_http, service_http_2784, service_http_443, service_http_8001, service_imap4, service_irc, service_iso_tsap, service_klogin, service_kshell, service_ldap, service_link, service_login, service_mtp, service_name, service_netbios_dgm, service_netbios_ns, service_netbios_ssn, service_netstat, service_nnsp, service_nntp, service_ntp_u, service_other, service_pm_dump, service_pop_2, service_pop_3, service_printer, service_private, service_red_i, service_remote_job, service_rje, service_shell, service_smtp, service_sql_net, service_ssh, service_sunrpc, service_supdup, service_systat, service_telnet, service_tftp_u, service_tim_i, service_time, service_urh_i, service_urp_i, service_uucp, service_uucp_path, service_vmnet, service_whois, service_x11, service_z39_50, flag_oth, flag_rej, flag_rsto, flag_rstos0, flag_rstr, flag_s0, flag_s1, flag_s2, flag_s3, flag_sf, flag_sh

Métriques finales (affichage UI / test set)

Métrique	Valeur
accuracy	0.766501
precision	0.807408
recall	0.766501
f1	0.731538
train_accuracy	0.992903
final_test.accuracy	0.766501
final_test.precision	0.807408
final_test.recall	0.766501
final_test.f1	0.731538
final_train.accuracy	0.992903
n_classes	5
classes	['0.0', '1.0', '2.0', '3.0', '4.0']
target_column	class
n_support_vectors	5393

Matrice de confusion (test)

actual \ pred	0.0	1.0	2.0	3.0	4.0
0.0	6279	1154	27	0	0
1.0	293	9021	391	1	5
2.0	349	387	1685	0	0
3.0	2	2583	15	282	3
4.0	0	37	13	4	13

Classification report (test)

Classe / avg	precision	recall	f1-score	support
0.0	0.906977	0.841689	0.873114	7460.0
1.0	0.684342	0.928947	0.788101	9711.0
2.0	0.790709	0.695993	0.740334	2421.0
3.0	0.982578	0.097747	0.177806	2885.0
4.0	0.619048	0.194030	0.295455	67.0
accuracy			0.766501	
macro avg	0.796731	0.551681	0.574962	22544.0
weighted avg	0.807408	0.766501	0.731538	22544.0

Feature importance (top 20)

Rang	Feature	Importance
1	hot	1464.678472
2	is_guest_login	1236.899419
3	service_ftp	1021.050264
4	service_ftp_data	710.875742
5	su_attempted	624.257416
6	dst_host_same_src_port_rate	561.815144
7	root_shell	476.308749
8	num_file_creations	417.137197
9	num_root	382.544840
10	num_access_files	368.607253
11	num_compromised	366.877657
12	urgent	347.681874
13	logged_in	347.229525
14	num_failed_logins	318.748998
15	service_telnet	310.686871
16	num_shells	301.769223
17	dst_host_count	299.628242
18	flag_sf	293.202187
19	same_srv_rate	289.260342
20	src_bytes	284.151515

training_info


```
{
  "final_training_data": "train_only",
  "all_data_samples": 125973,
  "train_samples_used": 125973,
  "test_samples_used": 0,
  "final_model_trained_on": "train_data_only",
  "n_repetitions": 5,
  "validation_method": "holdout",
  "date_column_excluded": false,
  "final_training_on_all_data": false,
  "training_samples": 125973,
```

```
  "test_samples": 22544,
  "final_model_samples": 125973,
  "feature_scaling": "StandardScaler applied",
  "parameter_optimization": "User parameters used - no optimization",
  "final_model_source": "best_validation_model"
}
```


Synthèse finale

Meilleure accuracy test: Support Vector Machine (SVM) (0.7665)

Plus rapide: LightGBM (17.5s)

Plus lent: Support Vector Machine (SVM) (4710.2s)

Logs bruts serveur: /root/tfmv1/backend/training_logs/